Bio-Ball: A Bio-Inspired Optimizer For Football Formations Across Three Phases Of Play

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***Abstract*—This work proposes a bio-inspired approach for optimizing football team formations across different phases of play. By formulating player positioning as a continuous black-box optimization problem and applying CMA-ES, realistic and tactically coherent formations are obtained, outperforming Differential Evolution used as a benchmark.**

I. Introduction

In recent years, football has become an increasingly dynamic sport, with teams capable of changing their tactical setup within the same match, for example shifting from a 4-4-2 formation to a 3-5-2 in order to adapt to different game situations.

The objective of this work was to develop an optimizer based on bio-inspired artificial intelligence techniques, specifically the **CMA-ES** (**Covariance Matrix Adaptation Evolution Strategy**) algorithm, capable of optimizing players positions across three main phases of play: **Offensive Possession**, when the team is in possession of the ball in the opponent's half; **Defensive Possession**, when possession occurs in the team's own half; and **Defensive Phase**, when the team is not in possession of the ball and is positioned in its own half.

II. Background Theory

Before describing the proposed methodology, this section briefly introduces the theoretical background of the optimization algorithms employed in this work.

A. CMA-ES (Covariance Matrix Adaptation Evolution Strategy)

CMA-ES is an evolutionary optimization algorithm belonging to the family of **Evolution Strategies**, particularly suited for continuous **black-box optimization problems**. It is designed to handle non-linear and non-convex objective functions without requiring gradient information.

The algorithm operates by iteratively sampling a population of candidate solutions, evaluating them through a fitness function, and updating its search strategy based on the best-performing individuals. Candidate solutions are generated from a multivariate Gaussian distribution, whose parameters are progressively adapted during the optimization process.

The effectiveness of CMA-ES mainly derives from the **adaptive update of its global search parameters**, specifically the step-size and the covariance matrix. By adapting the covariance matrix, the algorithm is able to capture dependencies between variables and to shape the search distribution according to the structure of the fitness landscape, leading to efficient exploration and robust convergence.

B. Differential Evolution

Differential Evolution (DE) is a population-based evolutionary optimization algorithm designed for continuous **black-box problems**. Unlike CMA-ES, DE does not rely on an adaptive probabilistic model, but directly exploits the geometric relationships between individuals in the population to guide the search.

As in other evolutionary approaches, DE aims to minimize a fitness function through an iterative process based on three main operators: mutation, crossover, and selection. New candidate solutions are generated by combining existing population members, encouraging exploration of the solution space, while a greedy selection mechanism ensures the retention of better-performing individuals.

Due to its simplicity and robustness, Differential Evolution is often used as a baseline method for comparison with more advanced optimization strategies.

III. Design Choises

A. Data Acquisition and Preprocessing

The work was structured starting from the data acquisition phase. To this end, the datasets provided by *Metrica Sports* were used. These consist of three main files: one file containing **match events**, and two separate files—one for the *home* team and one for the *away* team—containing the **positions of players** and the **ball** over time.

The events file was used to determine ball possession and to identify the current phase of play, while the tracking files were employed to reconstruct the spatial arrangement of players on the field.

B. Initial Population Generation

Subsequently, the initial populations provided to the optimization algorithms were constructed. As described in the introduction, the problem addressed involves the optimization of team formations across three different phases of play; therefore, an initial population was computed for each phase.

Each individual in the population is represented by 22 real-valued variables, corresponding to the (x, y) coordinates of the players of the considered team. The initial populations were obtained by analyzing the events file to determine ball possession and the half of the field in which play was occurring, using the ball position as a reference. For each phase of play, player positions were collected and then averaged over time, resulting in an initial configuration representative of the team's actual spatial arrangement.

This procedure allows the optimization algorithms to start from a realistic initial point, reducing the distance between the initial solution and tactically plausible configurations.

C. Fitness Function

The **fitness function** represents one of the core components of the proposed system. It was defined as a **weighted sum** of multiple **tactical objectives** and **physical constraints**, with the goal of guiding the optimization process toward solutions that are both tactically effective and spatially realistic.

Formally, the fitness function can be interpreted as the combination of two main components: a **penalty component**, which enforces physical and structural constraints, and a **tactical objective component**, which models the desired team behavior across the different phases of play.

*1) **Constraints**:* The constraints included in the fitness function are designed to prevent invalid or unrealistic solutions. In particular:

- **Boundary constraint**: penalizes solutions in which one or more players are located outside the boundaries of the playing field, ensuring the physical validity of the configurations.
- **Goalkeeper constraint**: enforces the goalkeeper to remain within a predefined allowed area. The penalty increases with the distance from the admissible region, preventing unrealistic use of the goalkeeper.
- **Collision and proximity constraint**: penalizes pairs of players whose distance falls below a minimum threshold, preventing collisions and unrealistic spatial overlaps.
- **Transition smoothness**: penalizes large positional changes between configurations belonging to consecutive phases of play, promoting smooth transitions and overall tactical coherence.

*2) **Tactical Objectives**:* The tactical objectives vary depending on the phase of play and were designed to model fundamental principles of modern football:

- **Field coverage (offensive phase)**: maximizes coverage of central and advanced areas of the field while penalizing less relevant regions such as defensive corners, encouraging a rational occupation of space.
- **Offside avoidance (offensive possession)**: penalizes players positioned beyond the offside line, which is dynamically defined based on the positions of the opponents and the ball.
- **Marking (defensive phase)**: ensures that each opponent, excluding the goalkeeper, is properly marked by penalizing large distances between defenders and opponents.
- **Defensive compactness**: minimizes player dispersion with respect to the team centroid, favoring a compact defensive structure.
- **Defensive line height**: dynamically regulates the height of the defensive line based on the ball position, penalizing both excessively deep lines and lines positioned beyond the ball.
- **Preventive marking**: penalizes unmarked opponents located in the team's own half during offensive phases, reducing the risk of counterattacks.
- **Ball pressure**: minimizes the distance between the closest player and the ball, modeling pressing behavior in defensive phases and support to the ball carrier in offensive phases.
- **Passing lanes availability**: evaluates the number and quality of available passing options, penalizing situations with limited options and rewarding configurations that promote effective combination play.

D. Static and Dynamic CMA-ES

With regard to optimization via CMA-ES, two different behaviors were implemented. In the **static** case, the *away* team maintains a fixed configuration throughout the entire optimization process, allowing the *home* team to adapt to an immutable scenario.

In the **dynamic** case, instead, the *away* team modifies its spatial arrangement during the optimization each generation through a heuristic and deterministic mechanism based on pressing and marking principles. This approach introduces increased realism into the simulated scenario, while not constituting a true co-evolutionary process, as the opponent team behavior is not optimized through a second evolutionary algorithm.

E. Differential Evolution

In addition to CMA-ES, **Differential Evolution** (DE) was employed as a baseline optimization algorithm for **benchmarking** purposes. DE was selected due to its simplicity, robustness, and widespread use in continuous black-box optimization, making it a suitable reference method for comparative evaluation. Within the project, DE operates on the same solution representation and fitness function used by CMA-ES, ensuring a fair comparison between the optimization strategies.

The primary role of Differential Evolution in this work is not to achieve the best absolute performance, but to serve as a benchmark for assessing the convergence behavior and solution quality of the proposed CMA-ES-based approaches under identical optimization settings.

F. Hyperparameter and Fitness Weight Optimization with Optuna

Optuna was employed not only for tuning the hyperparameters of the optimization algorithms, but also for **optimizing** the **weights of the fitness function** separately for each phase of play. In order to provide Optuna with a well-defined optimization target, an ideal reference formation was first constructed for each phase, encoding the desired tactical behavior in terms of player positioning.

During each Optuna trial, a specific set of algorithmic hyperparameters and fitness weights is sampled. The optimization process is then executed using these values, and the resulting formation is compared against the corresponding ideal reference formation. The discrepancy between the optimized formation and the ideal one is used as the objective value to be minimized by Optuna.

This formulation allows the hyperparameter search to be guided by a concrete tactical target, ensuring that the learned

fitness weights effectively encode phase-specific tactical principles while maintaining consistency with the underlying optimization framework.

IV. Experimental Setup

To obtain the final results, the optimization models were executed independently for each phase of play, producing an optimized formation for each scenario. All experiments were conducted using the same optimization settings across phases in order to ensure a fair and consistent comparison.

A. CMA-ES Configuration

Both the static and dynamic variants of CMA-ES were executed using identical hyperparameter configurations. The maximum number of iterations, population size, and initial step-size were fixed as follows:

- `CMA_MAX_ITER` = 160
- `CMA_POP_SIZE` = 20
- `CMA_SIGMA_INIT` = 0.15

These values were selected based on preliminary experiments and further refined through the hyperparameter optimization procedure described in the previous section.

B. Differential Evolution Configuration

Differential Evolution was employed as a benchmark optimizer and executed using the following configuration:

- `DE_MAX_ITER` = 50
- `DE_POP_SIZE` = 20
- `DE_MUTATION` = $(0.5, 1.0)$
- `DE_RECOMBINATION` = 0.7

Given its role as a baseline method, DE was configured with a smaller number of iterations compared to CMA-ES, while maintaining a comparable population size to allow meaningful performance comparisons.

V. Results

In order to evaluate the performance of the proposed optimization approaches, the results were analyzed through both qualitative and quantitative means. In particular, the optimized **formations** obtained for each phase of play were visually inspected, and the **convergence** behavior of the fitness function was analyzed across generations.

A. CMA-ES Static Behavoir

Several independent runs were performed using CMA-ES in both static and dynamic configurations. Across all experiments, the static variant consistently produced homogeneous and stable results.

The static CMA-ES demonstrated the **best overall performance**, yielding tactically **plausible** formations with an **organic spatial distribution** of players across all phases of play. The optimized configurations effectively exploit the objectives encoded in the fitness function, resulting in balanced solutions in terms of coverage, compactness, and positional support.

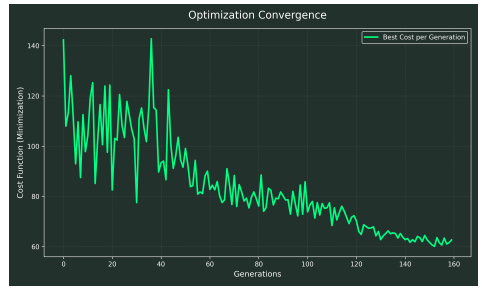

Fig. 1. Fitness convergence of static CMA-ES during the attacking possession phase.

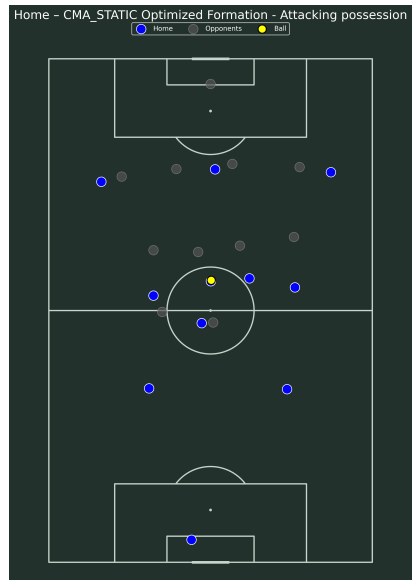

Fig. 2. Optimized formation obtained using static CMA-ES in attacking possession.

This qualitative assessment is further supported by the convergence plots. As shown in Figure 1, the fitness function exhibits an approximately monotonic decrease toward a minimum, indicating a stable and well-behaved optimization process.

An example of the optimized formation obtained using static CMA-ES in the attacking possession phase is shown in Figure 2. The resulting configuration presents a coherent team structure, with players occupying tactically meaningful positions relative to both teammates and opponents.

B. CMA-ES Dynamic Behavior

In contrast, the dynamic variant of CMA-ES resulted in significantly **poorer performance**. The optimized formations appear **unrealistic**, with players often converging toward locally optimal positions rather than contributing to a coherent collective structure.

This behavior is also reflected in the convergence plots. As shown in Figure 3, the fitness evolution is highly **noisy** and does not exhibit a clear convergence trend. This can be attributed to the non-stationary fitness landscape introduced by the dynamic opponent behavior, which changes at every

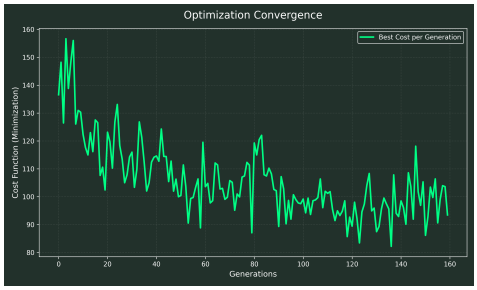

Fig. 3. Fitness convergence of dynamic CMA-ES during the attacking possession phase, showing unstable behavior.

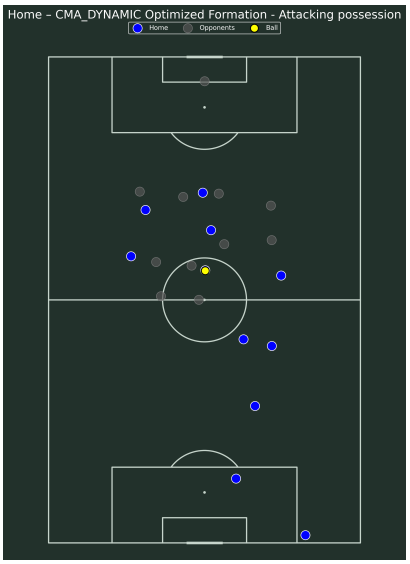

Fig. 4. Optimized formation obtained using dynamic CMA-ES in attacking possession.

generation and prevents the algorithm from learning a stable search distribution.

A representative optimized formation produced by the dynamic CMA-ES is shown in Figure 4. The lack of tactical coherence and excessive dispersion of players further confirm the limitations of this approach.

C. Differential Evolution Results

Differential Evolution was evaluated as a benchmark optimization method in order to assess the effectiveness of the proposed CMA-ES-based approach. The formations obtained using DE are generally characterized by **unrealistic** player distributions and a **lack of global tactical structure**.

As shown in Figure 5, DE fails to capture the strong spatial dependencies between players, resulting in scattered configurations that do not satisfy the tactical objectives encoded in the fitness function. These results further confirm that DE serves as a baseline method rather than a competitive alternative in this context.

VI. Conclusion

In this work, we demonstrated that bio-inspired optimization algorithms can be effectively employed to derive tactically

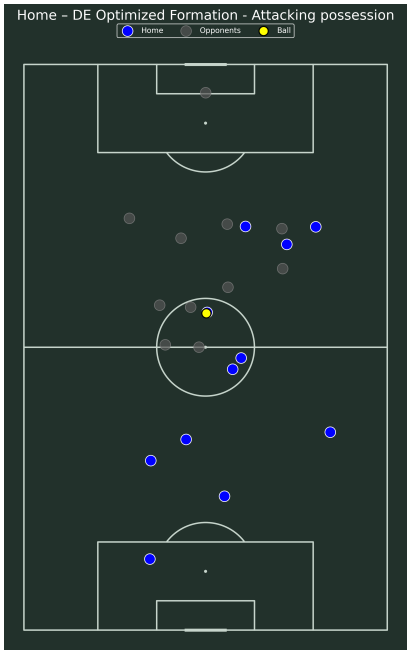

Fig. 5. Optimized formation obtained using Differential Evolution (benchmark) in attacking possession.

meaningful football formations. By modeling team positioning as a continuous optimization problem, the proposed approach was able to generate coherent and realistic formations across different phases of play.

Among the evaluated methods, static CMA-ES proved to be the most effective solution, consistently producing stable and tactically plausible formations. While this approach achieves high-quality results in terms of spatial organization, it does so at the cost of reduced realism in the modeling of the opponent behavior, which remains fixed during the optimization process. This highlights a fundamental trade-off between solution stability and opponent reactivity.

One of the main challenges addressed in this work was the mathematical formalization of tactical concepts within the fitness function. Translating qualitative football principles into quantitative objectives and constraints proved to be a critical and non-trivial aspect of the system design.

Future work could extend this framework by introducing multi-objective optimization techniques, allowing explicit trade-offs between competing tactical objectives. Additionally, incorporating player-specific attributes, such as preferred movement direction or speed, could further enhance the realism and expressiveness of the optimized formations.